Solution manual for Advanced Macroeconomics 5th Edition by David Romer

Intermediate Macroeconomics: Advanced (Harvard University)

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SOLUTIONS TO CHAPTER 1

Problem 1.1

(a) Since the growth rate of a variable equals the time derivative of its log, as shown by equation (1.10) in the text, we can write

$$(1)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{d\ln Z(t)}{dt} = \frac{d\ln[X(t)Y(t)]}{dt}.$$

Since the log of the product of two variables equals the sum of their logs, we have

$$(2)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{d[\ln X(t) + \ln Y(t)]}{dt} = \frac{d\ln X(t)}{dt} + \frac{d\ln Y(t)}{dt},$$

or simply

$$(3)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} + \frac{\dot{Y}(t)}{Y(t)}.$$

(b) Again, since the growth rate of a variable equals the time derivative of its log, we can write

$$(4)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{d\ln Z(t)}{dt} = \frac{d\ln[X(t)/Y(t)]}{dt}.$$

Since the log of the ratio of two variables equals the difference in their logs, we have

$$(5)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{d[\ln X(t) - \ln Y(t)]}{dt} = \frac{d\ln X(t)}{dt} - \frac{d\ln Y(t)}{dt},$$

or simply

$$(6)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{\dot{X}(t)}{X(t)} - \frac{\dot{Y}(t)}{Y(t)}.$$

(c) We have

$$(7)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{d\ln Z(t)}{dt} = \frac{d\ln[X(t)^{\alpha}]}{dt}.$$

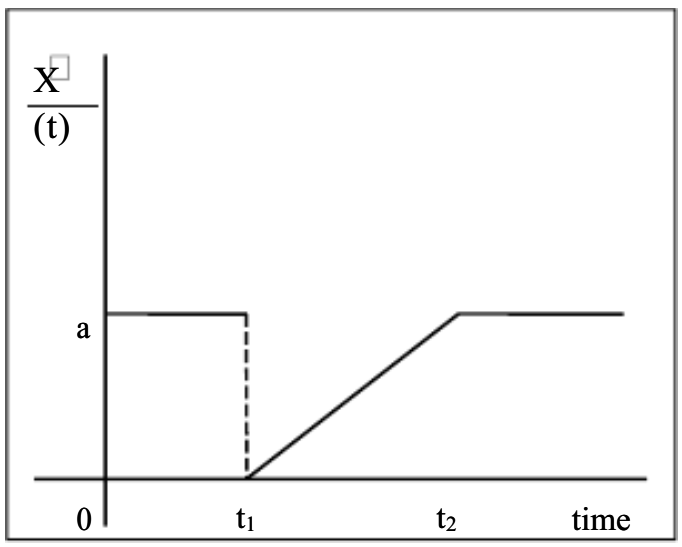
Using the fact that $\ln[X(t)^{\alpha}] = \alpha\ln X(t)$, we have

$$(8)\quad \frac{\dot{Z}(t)}{Z(t)} = \frac{d[\alpha\ln X(t)]}{dt} = \alpha\frac{d\ln X(t)}{dt} = \alpha\frac{\dot{X}(t)}{X(t)},$$

where we have used the fact that α is a constant.

Problem 1.2

(a) Using the information provided in the question, the path of the growth rate of X, $\dot{X}(t)/X(t)$, is depicted in the figure at right.



From time 0 to time t_1, the growth rate of X is constant and equal to a > 0. At time t_1, the growth rate of X drops to 0. From time t_1 to time t_2, the growth rate of X rises gradually from 0 to a. Note that we have made the assumption that $\dot{X}(t)/X(t)$ rises at a constant rate from t_1 to t_2. Finally, after time t_2, the growth rate of X is constant and equal to a again.

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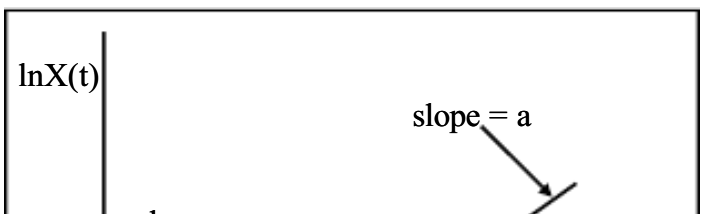
(b) Note that the slope of lnX(t) plotted against time is equal to the growth rate of X(t). That is, we know

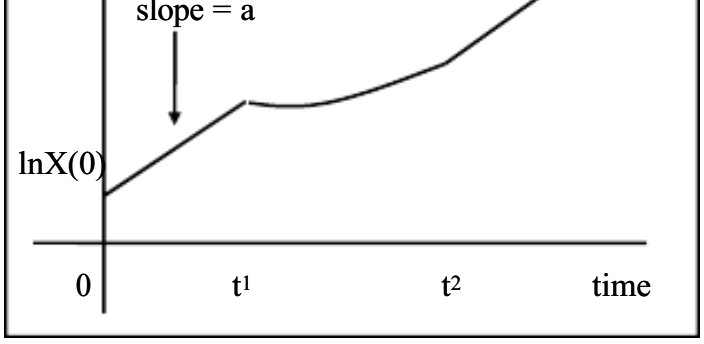
$$\frac{d \ln X(t)}{dt} = \frac{X(t)}{X(t)}$$

(See equation (1.10) in the text.)



(See equation (1.10) in the text.)

From time 0 to time t_1 the slope of lnX(t) equals $a > 0$. The lnX(t) locus has an inflection point at t_1, when the growth rate of X(t) changes discontinuously from a to 0. Between t_1 and t_2, the slope of lnX(t) rises gradually from 0 to a. After time t_2 the slope of lnX(t) is constant and equal to $a > 0$ again.

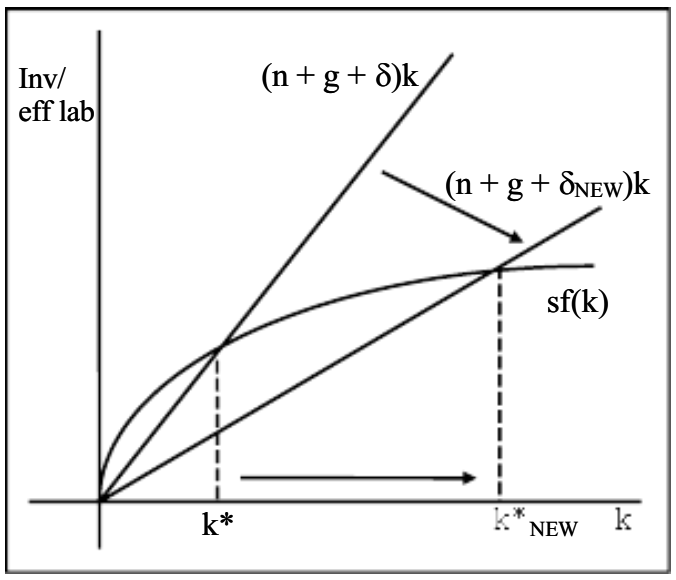


Problem 1.3

(a) The slope of the break-even investment line is given by $(n + g + \delta)$ and thus a fall in the rate of depreciation, δ, decreases the slope of the break-even investment line.

The actual investment curve, sf(k) is unaffected.

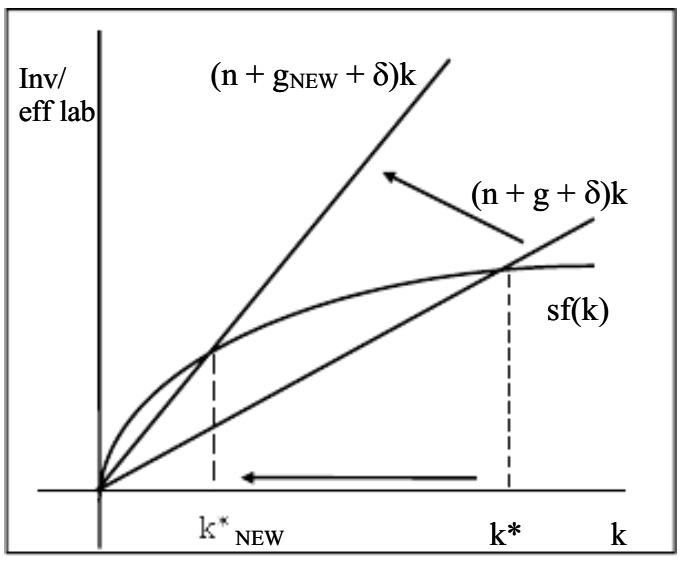
From the figure at right we can see that the balanced-growth-path level of capital per unit of effective labor rises from k^* to k^*_{NEW}.



(b) Since the slope of the break-even investment line is given by $(n + g + \delta)$, a rise in the rate of technological progress, g, makes the break-even investment line steeper.

The actual investment curve, sf(k), is unaffected.

From the figure at right we can see that the balanced-growth-path level of capital per unit of effective labor falls from k^* to k^*_{NEW}.



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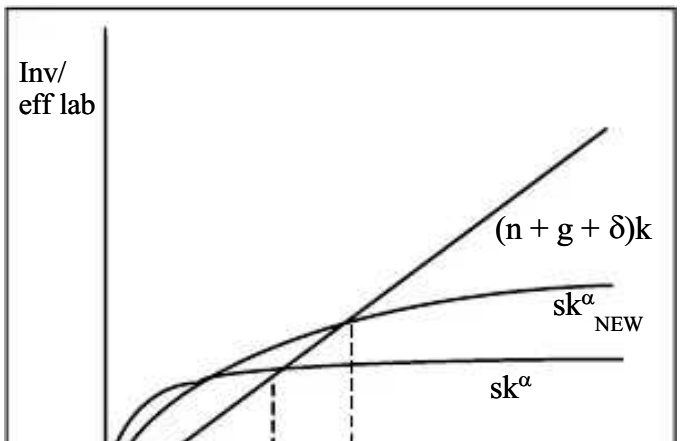
(c) The break-even investment line, $(n + g + \delta)k$, is unaffected by the rise in capital's share, α.

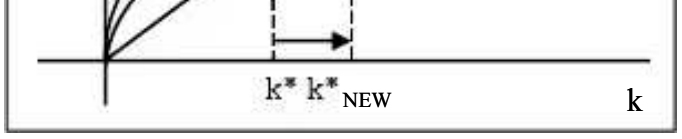
The effect of a change in α on the actual investment curve, sk^α, can be determined by examining the derivative $\partial(sk^\alpha)/\partial\alpha$. It is possible to show that

(1) $$\frac{\partial sk^\alpha}{\partial \alpha} = sk^\alpha \ln k .$$

For $0 < \alpha < 1$, and for positive values of k, the sign



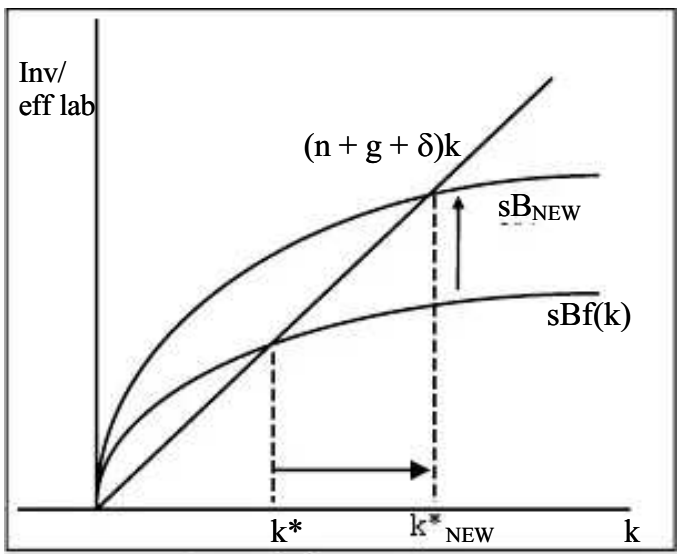
of $\partial(sk^{\alpha})/\partial\alpha$ is determined by the sign of lnk. For



lnk > 0, or k > 1, $\partial sk^{\alpha}/\partial\alpha > 0$ and so the new actual investment curve lies above the old one. For lnk < 0 or k < 1, $\partial sk^{\alpha}/\partial\alpha < 0$ and so the new actual investment curve lies below the old one. At k = 1, so that lnk = 0, the new actual investment curve intersects the old one.

In addition, the effect of a rise in α on k* is ambiguous and depends on the relative magnitudes of s and $(n + g + \delta)$. It is possible to show that a rise in capital's share, α, will cause k* to rise if $s > (n + g + \delta)$. This is the case depicted in the figure above.

(d) Suppose we modify the intensive form of the production function to include a non-negative constant, B, so that the actual investment curve is given by sBf(k), B > 0.



Then workers exerting more effort, so that output per unit of effective labor is higher than before, can be modeled as an increase in B. This increase in B shifts the actual investment curve up.

The break-even investment line, $(n + g + \delta)k$, is unaffected.

From the figure at right we can see that the balanced-growth-path level of capital per unit of effective labor rises from k* to k^*_{NEW} .

Problem 1.4

(a) At some time, call it t_0 , there is a discrete upward jump in the number of workers. This reduces the amount of capital per unit of effective labor from k* to k_{NEW} . We can see this by simply looking at the definition, $k \equiv K/AL$. An increase in L without a jump in K or A causes k to fall. Since f ' (k) > 0, this fall in the amount of capital per unit of effective labor reduces the amount of output per unit of effective labor as well. In the figure below, y falls from y* to y_{NEW} .

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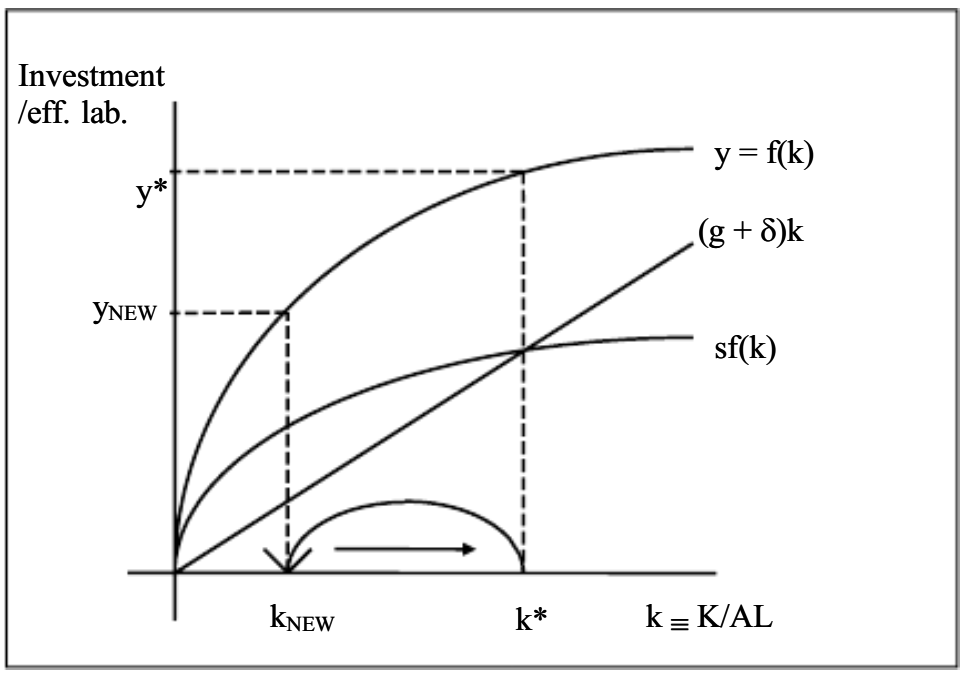
(b) Now at this lower k_{NEW} , actual investment per unit of effective labor exceeds break-even investment per unit of effective labor. That is, $sf(k_{NEW}) > (g + \delta)k_{NEW}$. The economy is now saving and investing more than enough to offset depreciation and technological progress at this lower k_{NEW} . Thus k begins rising back toward k*. As capital per unit of effective labor begins rising, so does output per unit of effective labor. That is, y begins rising from y_{NEW} back toward y*.



(c) Capital per unit of effective labor will continue to rise until it eventually returns to the original level

(c) Capital per unit of effective labor will continue to rise until it eventually returns to the original level of k*. At k*, investment per unit of effective labor is again just enough to offset technological progress and depreciation and keep k constant. Since k returns to its original value of k* once the economy again returns to a balanced growth path, output per unit of effective labor also returns to its original value of $y^* = f(k^*)$.

Problem 1.5

(a) The equation describing the evolution of the capital stock per unit of effective labor is given by

(1) $\dot{k} = sf(k) - (n + g + \delta)k$.

Substituting in for the intensive form of the Cobb-Douglas, $f(k) = k^{\alpha}$, yields

(2) $\dot{k} = sk^{\alpha} - (n + g + \delta)k$.

On the balanced growth path, $\dot{k}$ is zero; investment per unit of effective labor is equal to break-even investment per unit of effective labor and so k remains constant. Denoting the balanced-growth-path value of k as k*, we have $sk^{*\alpha} = (n + g + \delta)k^*$. Rearranging to solve for k* yields

(3) $k^* = \left[s/(n+g+\delta)\right]^{1/(1-\alpha)}$.

To get the balanced-growth-path value of output per unit of effective labor, substitute equation (3) into the intensive form of the production function, $y = k^{\alpha}$:

(4) $y^* = \left[s/(n+g+\delta)\right]^{\alpha/(1-\alpha)}$.

Consumption per unit of effective labor on the balanced growth path is given by $c^* = (1 - s)y^*$. Substituting equation (4) into this expression yields

(5) $c^* = (1-s)\left[s/(n+g+\delta)\right]^{\alpha/(1-\alpha)}$.

(b) The golden-rule level of the capital stock is that level at which consumption per unit of effective labor is maximized. To derive this level of k, take equation (3), which expresses the balanced-growth-path level of k, and rearrange it to solve for s:

(6) $s = (n + g + \delta)k^{*1-\alpha}$.

Now substitute equation (6) into equation (5):

(7) $c^* = \left[1-(n+g+\delta)k^{*1-\alpha}\right]\left[(n+g+\delta)k^{*1-\alpha}/(n+g+\delta)\right]^{\alpha/(1-\alpha)}$.

After some straightforward algebraic manipulation, this simplifies to

(8) $c^* = k^{*\alpha} - (n + g + \delta)k^*$.

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Equation (8) states that consumption per unit of effective labor is equal to output per unit of effective labor, $k^{*\alpha}$, less actual investment per unit of effective labor. On the balanced growth path, actual investment per unit of effective labor is the same as break-even investment per unit of effective labor, $(n + g + \delta)k^*$.

Now use equation (8) to maximize c* with respect to k*. The first-order condition is given by

(9) $\partial c^*/\partial k^* = \alpha k^{*\alpha-1} - (n+g+\delta) = 0$,

or simply

(10) $\alpha k^{*\alpha-1} = (n + g + \delta)$.

Note that equation (10) is just a specific form of the general condition that implicitly defines the golden-rule level of capital per unit of effective labor, given by $f'(k^*) = (n + g + \delta)$. Equation (10) has a graphical interpretation: it defines the level of k at which the slope of the intensive form of the production function is equal to the slope of the break-even investment line. Solving equation (10) for the golden-rule level of k yields

(11) $k^*_{GR} = \left[\alpha/(n+g+\delta)\right]^{1/(1-\alpha)}$.

(c) To get the saving rate that yields the golden-rule level of k, substitute equation (11) into (6):

(12) $s_{GR} = (n + g + \delta)\left[\alpha/(n+g+\delta)\right]^{(1-\alpha)/(1-\alpha)}$,

which simplifies to

(13) $s_{GR} = \alpha$.

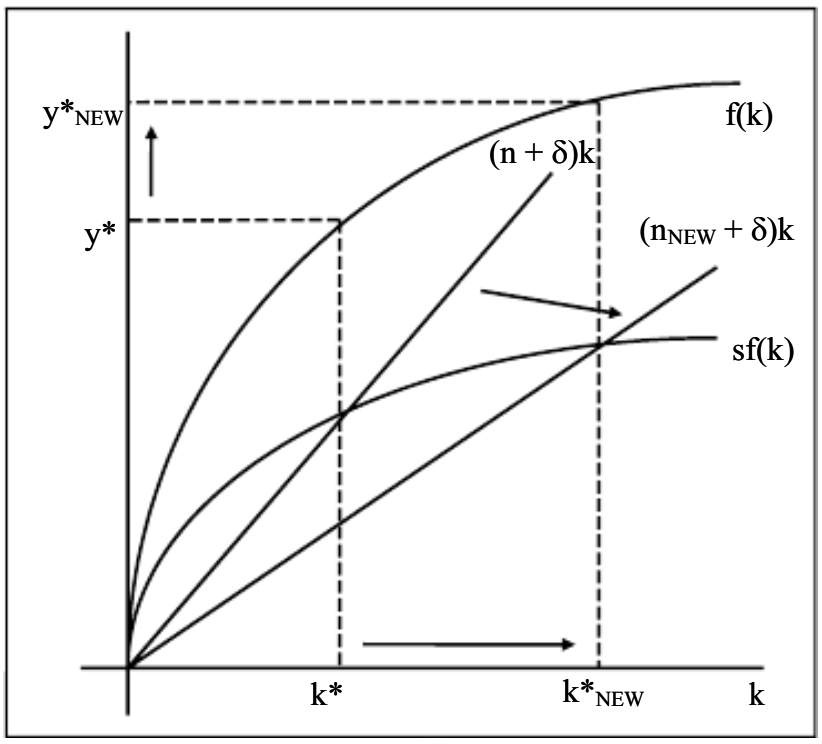
With a Cobb-Douglas production function, the saving rate required to reach the golden rule is equal to

the elasticity of output with respect to capital or capital's share in output (if capital earns its marginal product).

Problem 1.6

(a) Since there is no technological progress, we can carry out the entire analysis in terms of capital and output per worker rather than capital and output per unit of effective labor. With A constant, they behave the same. Thus we can define $y \equiv Y/L$ and $k \equiv K/L$.

The fall in the population growth rate makes the break-even investment line flatter. In the absence of technological progress, the per unit time change in k, capital per worker, is given by $\dot{k} = sf(k) - (\delta + n)k$. Since $\dot{k}$ was 0 before the decrease in n – the economy was on a balanced growth path – the decrease in n causes $\dot{k}$ to become positive. At k*, actual investment per worker, sf(k*), now exceeds break-even investment per worker, $(n_{NEW} + \delta)k^*$. Thus k moves to a new higher balanced growth path level. See the figure at right.



As k rises, y – output per worker – also rises. Since a constant fraction of output is saved, c – consumption per worker – rises as y rises. This is summarized in the figures below.

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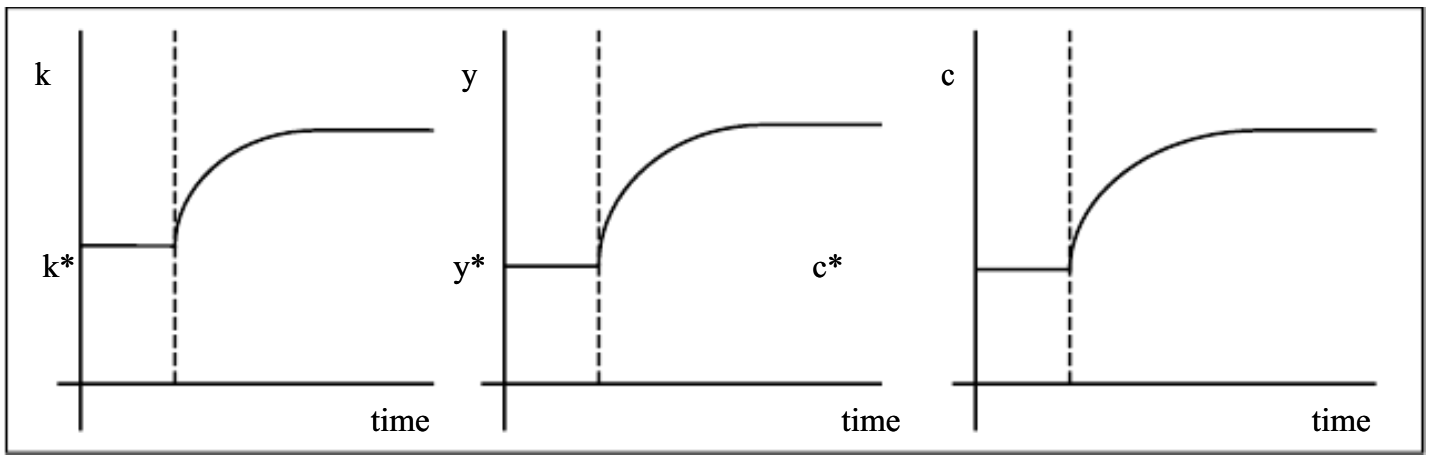
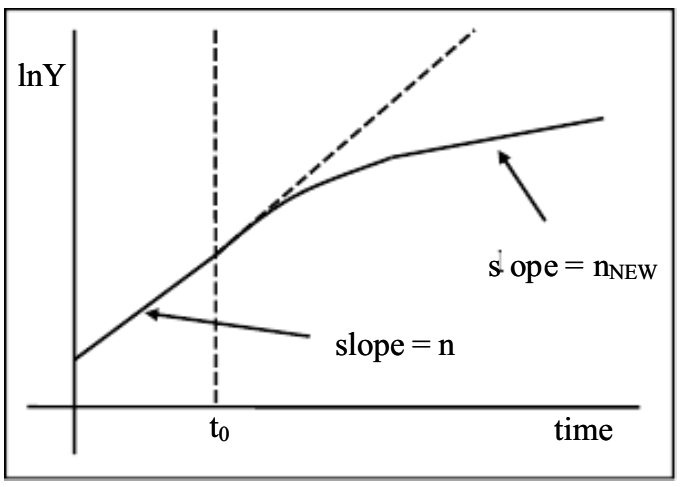


(b) By definition, output can be written as $Y \equiv Ly$. Thus the growth rate of output is $\dot{Y}/Y = \dot{L}/L + \dot{y}/y$. On the initial balanced growth path, $\dot{y}/y = 0$ – output per worker is constant – so $\dot{Y}/Y = \dot{L}/L = n$. On the final balanced growth path, $\dot{y}/y = 0$ again – output per worker is constant again – and so $\dot{Y}/Y = \dot{L}/L = n_{NEW} < n$. In the end, output will be growing at a permanently lower rate.



What happens during the transition? Examine the production function Y = F(K,AL). On the initial balanced growth path AL, K and thus Y are all growing at rate n. Then suddenly AL begins growing at some new lower rate n_{NEW}. Thus suddenly Y will be growing at some rate between that of K (which is

some new lower rate n_{NEW}. Thus suddenly Y will be growing at some rate between that of K (which is growing at n) and that of AL (which is growing at n_{NEW}). Thus, during the transition, output grows more rapidly than it will on the new balanced growth path, but less rapidly than it would have without the decrease in population growth. As output growth gradually slows down during the transition, so does capital growth until finally K, AL, and thus Y are all growing at the new lower n_{NEW}.

Problem 1.7

The derivative of $y^* = f(k^*)$ with respect to n is given by

(1) $\partial y^*/\partial n = f'(k^*)[\partial k^*/\partial n]$.

To find $\partial k^*/\partial n$, use the equation for the evolution of the capital stock per unit of effective labor, $\dot{k} = sf(k) - (n + g + \delta)k$. In addition, use the fact that on a balanced growth path, $\dot{k} = 0$, $k = k^*$ and thus $sf(k^*) = (n + g + \delta)k^*$. Taking the derivative of both sides of this expression with respect to n yields

(2) $$sf'(k^*)\frac{\partial k^*}{\partial n} = (n+g+\delta)\frac{\partial k^*}{\partial n} + k^*,$$

and rearranging yields

(3) $$\frac{\partial k^*}{\partial n} = \frac{k^*}{sf'(k^*) - (n+g+\delta)}.$$

Substituting equation (3) into equation (1) gives us

(4) $$\frac{\partial y^*}{\partial n} = f'(k^*)\left[\frac{k^*}{sf'(k^*) - (n+g+\delta)}\right].$$

Rearranging the condition that implicitly defines k^*, $sf(k^*) = (n + g + \delta)k^*$, and solving for s yields

(5) $s = (n + g + \delta)k^*/f(k^*)$.

Substitute equation (5) into equation (4):

(6) $$\frac{\partial y^*}{\partial n} = \frac{f'(k^*)k^*}{[(n+g+\delta)f'(k^*)k^*/f(k^*)] - (n+g+\delta)}.$$

To turn this into the elasticity that we want, multiply both sides of equation (6) by n/y^*:

(7) $$\frac{n}{y^*}\frac{\partial y^*}{\partial n} = \frac{n}{(n+g+\delta)}\frac{f'(k^*)k^*/f(k^*)}{[f'(k^*)k^*/f(k^*)] - 1}.$$

Using the definition that $\alpha_K(k^*) \equiv f'(k^*)k^*/f(k^*)$ gives us

(8) $$\frac{n}{y^*}\frac{\partial y^*}{\partial n} = -\frac{n}{(n+g+\delta)}\left[\frac{\alpha_K(k^*)}{1 - \alpha_K(k^*)}\right].$$

Now, with $\alpha_K(k^*) = 1/3$, $g = 2\%$ and $\delta = 3\%$, we need to calculate the effect on y^* of a fall in n from 2% to 1%. Using the midpoint of $n = 0.015$ to calculate the elasticity gives us

(9) $$\frac{n}{y^*}\frac{\partial y^*}{\partial n} = -\frac{0.015}{(0.015 + 0.02 + 0.03)}\left(\frac{1/3}{1 - 1/3}\right) \cong -0.12.$$

So this 50% drop in the population growth rate, from 2% to 1%, will lead to approximately a 6% increase in the level of output per unit of effective labor, since (-0.50)(-0.12) = 0.06. This calculation illustrates the point that observed differences in population growth rates across countries are not nearly enough to account for differences in y that we see.

Problem 1.8

(a) A permanent increase in the fraction of output that is devoted to investment from 0.15 to 0.18 represents a 20 percent increase in the saving rate. From equation (1.28) in the text, the elasticity of output with respect to the saving rate is

(1) $$\frac{s}{y^*}\frac{\partial y^*}{\partial s} = \frac{\alpha_K(k^*)}{1 - \alpha_K(k^*)},$$

where $\alpha_K(k^*)$ is the share of income paid to capital (assuming that capital is paid its marginal product).

Substituting the assumption that $\alpha_K(k^*) = 1/3$ into equation (1) gives us

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Substituting the assumption that $\alpha_K(k^*) = 1/3$ into equation (1) gives us

(2) $$\frac{s}{y^*}\frac{\partial y^*}{\partial s} = \frac{\alpha_K(k^*)}{1-\alpha_K(k^*)} = \frac{1/3}{1-1/3} = \frac{1}{2}.$$

Thus the elasticity of output with respect to the saving rate is 1/2. So this 20 percent increase in the saving rate – from s = 0.15 to $s_{NEW} = 0.18$ – causes output to rise relative to what it would have been by about 10 percent. [Note that the analysis has been carried out in terms of output per unit of effective labor. Since the paths of A and L are not affected, however, if output per unit of effective labor rises by 10 percent, output itself is also 10 percent higher than what it would have been.]

(b) Consumption rises less than output. Output ends up 10 percent higher than what it would have been. But the fact that the saving rate is higher means that we are now consuming a smaller fraction of output. We can calculate the elasticity of consumption with respect to the saving rate. On the balanced growth path, consumption is given by

(3) $c^* = (1 - s)y^*$.

Taking the derivative with respect to s yields

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(4) $$\frac{\partial c^*}{\partial s} = -y^* + (1-s)\frac{\partial y^*}{\partial s}.$$

To turn this into an elasticity, multiply both sides of equation (4) by s/c*:

(5) $$\frac{\partial c^*}{\partial s}\frac{s}{c^*} = \frac{-y^* s}{(1-s)y^*} + (1-s)\frac{\partial y^*}{\partial s}\frac{s}{(1-s)y^*},$$

where we have substituted $c^* = (1 - s)y^*$ on the right-hand side. Simplifying gives us

(6) $$\frac{\partial c^*}{\partial s}\frac{s}{c^*} = \frac{-s}{(1-s)} + \frac{\partial y^*}{\partial s}\frac{s}{(1-s)y^*}.$$

From part (a), the second term on the right-hand side of (6), the elasticity of output with respect to the saving rate, equals 1/2. We can use the midpoint between s = 0.15 and $s_{NEW} = 0.18$ to calculate the elasticity:

(7) $$\frac{\partial c^*}{\partial s}\frac{s}{c^*} = \frac{-0.165}{(1-0.165)} + 0.5 \cong 0.30.$$

Thus the elasticity of consumption with respect to the saving rate is approximately 0.3. So this 20% increase in the saving rate will cause consumption to be approximately 6% above what it would have been.

(c) The immediate effect of the rise in investment as a fraction of output is that consumption falls. Although y* does not jump immediately – it only begins to move toward its new, higher balanced-growth-path level – we are now saving a greater fraction, and thus consuming a smaller fraction, of this same y*. At the moment of the rise in s by 3 percentage points – since $c = (1 - s)y^*$ and y* is unchanged – c falls. In fact, the percentage change in c will be the percentage change in (1 - s). Now, (1 - s) falls from 0.85 to 0.82, which is approximately a 3.5 percent drop. Thus at the moment of the rise in s, consumption falls by about three and a half percent.

We can use some results from the text on the speed of convergence to determine the length of time it takes for consumption to return to what it would have been without the increase in the saving rate. After the initial rise in s, s remains constant throughout. Since $c = (1 - s)y$, this means that consumption will grow at the same rate as y on the way to the new balanced growth path. In the text it is shown that the rate of convergence of k and y, after a linear approximation, is given by $\lambda = (1 - \alpha_K)(n + g + \delta)$. With $(n + g + \delta)$ equal to 6 percent per year and $\alpha_K = 1/3$, this yields a value for λ of about 4 percent. This means that k and y move about 4 percent of the remaining distance toward their balanced-growth-path values of k* and y* each year. Since c is proportional to y, $c = (1 - s)y$, it also approaches its new balanced-growth-path value at that same constant rate. That is, analogous to equation (1.31) in the text, we could write

(8) $$c(t) - c^* \cong e^{-(1-\alpha_K)(n+g+\delta)t}[c(0) - c^*],$$

or equivalently

(9) $e^{-\lambda t} = \dfrac{c(t) - c^*}{c(0) - c^*}$.

The term on the right-hand side of equation (9) is the fraction of the distance to the balanced growth path that remains to be traveled.

We know that consumption falls initially by 3.5 percent and will eventually be 6 percent higher than it would have been. Thus it must change by 9.5 percent on the way to the balanced growth path. It will therefore be equal to what it would have been about 36.8 percent (3.5%/9.5% ≅ 36.8%) of the way to the new balanced growth path. Equivalently, this is when the remaining distance to the new balanced growth

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path is 63.2 percent of the original distance. To determine the length of time this will take, we need to find a t* that solves

(10) $e^{-\lambda t^*} = 0.632$.

Taking the natural logarithm of both sides of equation (10) yields

(11) $-\lambda t^* = \ln(0.632)$.

Rearranging to solve for t gives us

(12) $t^* = 0.459/0.04$,

and thus

(13) $t^* \cong 11.5$ years.

It will take a long time – over a decade – for consumption to return to what it would have been in the absence of the increase in investment as a fraction of output.

Problem 1.9

(a) Define the marginal product of labor to be $w \equiv \partial F(K,AL)/\partial L$. Then write the production function as $Y = ALf(k) = ALf(K/AL)$. Taking the partial derivative of output with respect to L yields

(1) $w \equiv \partial Y/\partial L = ALf'(k)[-K/AL^2] + Af(k) = A[(-K/AL)f'(k) + f(k)] = A[f(k) - kf'(k)]$,

as required.

(b) Define the marginal product of capital as $r \equiv [\partial F(K,AL)/\partial K] - \delta$. Again, writing the production function as $Y = ALf(k) = ALf(K/AL)$ and now taking the partial derivative of output with respect to K yields

(2) $r \equiv [\partial Y/\partial K] - \delta = ALf'(k)[1/AL] - \delta = f'(k) - \delta$.

Substitute equations (1) and (2) into wL + rK:

(3) $wL + rK = A[f(k) - kf'(k)]L + [f'(k) - \delta]K = ALf(k) - f'(k)[K/AL]AL + f'(k)K - \delta K$.

Simplifying gives us

(4) $wL + rK = ALf(k) - f'(k)K + f'(k)K - \delta K = Alf(k) - \delta K \equiv ALF(K/AL, 1) - \delta K$.

Finally, since F is constant returns to scale, equation (4) can be rewritten as

(5) $wL + rK = F(ALK/AL, AL) - \delta K = F(K, AL) - \delta K$.

(c) As shown above, $r = f'(k) - \delta$. Since δ is a constant and since k is constant on a balanced growth path, so is f'(k) and thus so is r. In other words, on a balanced growth path, $\partial r/\partial r = 0$. Thus the Solow model does exhibit the property that the return to capital is constant over time.

Since capital is paid its marginal product, the share of output going to capital is rK/Y. On a balanced growth path,

(6) $\dfrac{\partial(rK/Y)}{(rK/Y)} = \dfrac{\partial r}{r} + \dfrac{\partial K}{K} - \dfrac{\partial Y}{Y} = 0 + (n + g) - (n + g) = 0$.

Thus, on a balanced growth path, the share of output going to capital is constant. Since the shares of output going to capital and labor sum to one, this implies that the share of output going to labor is also constant on the balanced growth path.

We need to determine the growth rate of the marginal product of labor, w, on a balanced growth path. As

shown above, $w = A[f(k) - kf'(k)]$. Taking the time derivative of the log of this expression yields the growth rate of the marginal product of labor:

$$(7)\quad \frac{\dot{w}}{w} = \frac{\dot{A}}{A} + \frac{[f(k) - kf'(k)]^{\cdot}}{[f(k) - kf'(k)]} = g + \frac{[f'(k)\dot{k} - \dot{k}f'(k) - kf''(k)\dot{k}]}{f(k) - kf'(k)} = g + \frac{-kf''(k)\dot{k}}{f(k) - kf'(k)}.$$

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On a balanced growth path $\dot{k} = 0$ and so $\dot{w}/w = g$. That is, on a balanced growth path, the marginal product of labor rises at the rate of growth of the effectiveness of labor.

(d) As shown in part (c), the growth rate of the marginal product of labor is

$$(8)\quad \frac{\dot{w}}{w} = g + \frac{-kf''(k)\dot{k}}{f(k) - kf'(k)}.$$

If $k < k^*$, then as k moves toward k^*, $\dot{w}/w > g$. This is true because the denominator of the second term on the right-hand side of equation (8) is positive because f(k) is a concave function. The numerator of that same term is positive because k and $\dot{k}$ are positive and $f''(k)$ is negative. Thus, as k rises toward k^*, the marginal product of labor grows faster than on the balanced growth path. Intuitively, the marginal product of labor rises by the rate of growth of the effectiveness of labor on the balanced growth path. As we move from k to k^*, however, the amount of capital per unit of effective labor is also rising which also makes labor more productive and this increases the marginal product of labor even more.

The growth rate of the marginal product of capital, r, is

$$(9)\quad \frac{\dot{r}}{r} = \frac{[f'(k)]^{\cdot}}{f'(k)} = \frac{f''(k)\dot{k}}{f'(k)}.$$

As k rises toward k^*, this growth rate is negative since $f'(k) > 0$, $f''(k) < 0$ and $\dot{k} > 0$. Thus, as the economy moves from k to k^*, the marginal product of capital falls. That is, it grows at a rate less than on the balanced growth path where its growth rate is 0.

Problem 1.10

(a) The growth rate of the capital-output ratio is given by

$$(1)\quad \frac{(K/Y)^{\cdot}}{(K/Y)} = \frac{\dot{K}}{K} - \frac{\dot{Y}}{Y}.$$

Differentiating the production function, $Y = F(K,AL)$, with respect to time gives us

$$(2)\quad \dot{Y} = F_K\dot{K} + F_{AL}(AL)^{\cdot}.$$

Dividing both sides of equation (2) by Y yields

$$(3)\quad \frac{\dot{Y}}{Y} = \frac{F_K K}{Y}\frac{\dot{K}}{K} + \frac{ALF_{AL}}{Y}\frac{(AL)^{\cdot}}{AL}.$$

Defining $\alpha_K \equiv F_K K/Y$, which is capital's share of income, and using the fact that the production function exhibits constant returns to scale allows us to write

$$(4)\quad \frac{\dot{Y}}{Y} = \alpha_K\frac{\dot{K}}{K} + (1-\alpha_K)\frac{(AL)^{\cdot}}{AL}.$$

Since A grows at rate g and L grows at rate n, this becomes

$$(5)\quad \frac{\dot{Y}}{Y} = \alpha_K\frac{\dot{K}}{K} + (1-\alpha_K)(n+g).$$

Substituting equation (5) into equation (1) yields

$$(6)\quad \frac{(K/Y)^{\cdot}}{(K/Y)} = \frac{\dot{K}}{K} - \alpha_K\frac{\dot{K}}{K} - (1-\alpha_K)(n+g) = (1-\alpha_K)\left[\frac{\dot{K}}{K} - (n+g)\right].$$

$$(K/Y) \quad K \quad K \quad K \quad \left[K \quad \right]$$

Thus, the growth rate of the capital-output ratio is positive, i.e. K/Y is rising, if and only if the growth rate of the capital stock exceeds n + g. Since $k \equiv K/AL$, when K is growing faster than the growth rate of AL, which is n + g, k is rising. So, equivalently, K/Y is rising if and only if k is rising.

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(b) The solution to part (a) shows us that we can examine the behavior of K/Y by looking at the behavior of k. The equation describing the evolution of the capital stock per unit of effective labor in the standard version of the Solow model is given by

(7) $\dot{k} = sf(k) - (n + g + \delta)k$.

The assumption here that capital's share of income is saved implies that $s = F_K K/Y$. Since $F_K = f'(k)$, this implies $s = f'(k)k/f(k)$. Thus, equation (7) can be written as

(8) $\dot{k} = f'(k)k - (n + g + \delta)k$.

Dividing both sides of (8) by k gives us an expression for the growth rate of capital per unit of effective labor,

(9) $\dfrac{\dot{k}}{k} = f'(k) - (n + g + \delta)$.

From equation (9), we can see that if the marginal product of capital, which equals f′(k), is strictly greater than (n + g + δ) at time t = 0, $\dot{k}/k$ will be positive and so k will rise over time. As shown in the solution to part (a), if k is rising so is K/Y. Thus, K/Y rises over time. But since $f''(k) < 0$, f′(k) falls as k rises. Eventually f′(k) will fall to (n + g + δ), at which point k becomes constant. The solution to part (a) implies that K/Y will also become constant at that point. In summary, K/Y gradually approaches a constant value from below.

(c) The solution to part (b) shows that K/Y does not grow without bound. The key is that the return to capital is decreasing in k. As capital accumulates, the return to capital falls, and so capital accumulates more slowly. Eventually, the return to capital falls so much that capital income, and thus savings, is just enough to offset depreciation, population growth, and growth in efficiency and so K/Y becomes constant.

(d) With a Cobb-Douglas production function, $Y = K^{\alpha}(AL)^{1-\alpha}$, the marginal product of capital is given by

(10) $\dfrac{\partial Y}{\partial K} = \alpha K^{\alpha-1}(AL)^{1-\alpha}$.

Thus, we can write net-capital income, NCI, as

(11) $NCI = K\left[\dfrac{\partial Y}{\partial K} - \delta\right] = K[\alpha K^{\alpha-1}(AL)^{1-\alpha} - \delta] = \alpha Y - \delta K$.

Taking the ratio of net capital income to net output, NI, gives us

(12) $\dfrac{NCI}{NI} = \dfrac{\alpha Y - \delta K}{Y - \delta K}$.

Dividing the top and bottom of the right-hand-side of equation (12) by Y yields

(13) $\dfrac{NCI}{NI} = \dfrac{\alpha - \delta K/Y}{1 - \delta K/Y}$.

Differentiating the ratio of net capital income to net output with respect to the capital-output ratio, K/Y, gives us

(14) $\dfrac{\partial NCI/NI}{\partial K/Y} = \dfrac{[-\delta](1 - \delta K/Y) - (-\alpha - \delta K/Y)[-\delta]}{[1 - \delta K/Y]^2}$,

which simplifies to

(15) $\dfrac{\partial NCI/NI}{\partial K/Y} = \dfrac{-\delta + \delta^2 K/Y + \alpha\delta - \delta^2 K/Y}{[1 - \delta K/Y]^2} = \dfrac{\delta(\alpha - 1)}{[1 - \delta K/Y]^2} < 0$.

Equation (15) implies that as K/Y rises over time and gradually approaches a constant level, the ratio of net capital income to net output is falling. So the common statement that an excess of the return to capital over the economy's growth rate causes capital's share to rise over time is not correct in this case.

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Problem 1.11

(a) The elasticity of substitution between capital and effective labor is given by

(1) $$\sigma \equiv \frac{\partial \log(K/AL)}{\partial \log(F_{AL}/F_K)} = -\frac{\partial \log(K/AL)}{\partial \log(F_K/F_{AL})}.$$

This can be rewritten as

(2) $$\sigma = -\frac{\partial \log(K/Y)}{\partial \log(F_K)}.$$

The share of net capital income in net output is given by

(3) $$\frac{NCI}{NI} = \frac{K(F_K - \delta)}{Y - \delta K} = \frac{F_K - \delta}{Y/K - \delta}.$$

To see the direction of the response of NCI/NI to changes in K, we can examine the derivative of NCI/NI with respect to F_K since we know that $\partial F_K / \partial K < 0$. We have

(4) $$\frac{\partial(NCI/NI)}{\partial F_K} = \frac{[1](Y/K - \delta) - (F_K - \delta)[\partial(Y/K)/\partial F_K]}{[Y/K - \delta]^2}.$$

Equation (4) simplifies to

(5) $$\frac{\partial(NCI/NI)}{\partial F_K} = \frac{1}{Y/K - \delta} - \frac{(F_K - \delta)}{[Y/K - \delta]^2}\frac{\partial(Y/K)}{\partial F_K}.$$

From equation (2), we have

(6) $$\sigma = \frac{\partial \log(Y/K)}{\partial \log(F_K)} = \frac{\partial(Y/K)}{\partial F_K}\frac{(K/Y)}{F_K}.$$

Solving for $\partial(Y/K)/\partial F_K$ gives us

(7) $$\frac{\partial(Y/K)}{\partial F_K} = \sigma\frac{(Y/K)}{F_K}.$$

Substituting equation (7) into equation (5) yields

(8) $$\frac{\partial(NCI/NI)}{\partial F_K} = \frac{1}{Y/K - \delta} - \sigma\frac{(F_K - \delta)}{[Y/K - \delta]^2}\frac{Y/K}{F_K}.$$

Thus, a marginal increase in K will increase the share of net capital income in net output if this expression is negative.

(b) We require a solution for σ such that

(9) $$\frac{1}{Y/K - \delta} - \sigma\frac{(F_K - \delta)}{[Y/K - \delta]^2}\frac{Y/K}{F_K} < 0.$$

Substituting in the parameter values that we are given yields

(10) $$\frac{1}{1/3 - 0.03} - \sigma\frac{0.05}{[1/3 - 0.03]^2}\frac{1/3}{0.08} < 0,$$

or simply

(11) $$\sigma > \frac{\frac{1}{1/3 - 0.03}}{\frac{0.05}{[1/3 - 0.03]^2}\frac{1/3}{0.08}} = 1.456.$$

Thus, the elasticity of substitution must be greater than 1.456 for the share of net capital income in net output to rise when K rises.

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Problem 1.12

A balanced growth path occurs when all the variables of the model are growing at constant rates. Despite the differences between this model and the usual Solow model, it turns out that we can again show that the economy will converge to a balanced growth path by examining the behavior of $k \equiv K/AL$.

Taking the time derivative of both sides of the definition of $k \equiv K/AL$ gives us

$$(1)\quad \dot{k} = \dot{\left(\frac{K}{AL}\right)} = \frac{\dot{K}(AL) - K\left[\dot{L}A - A\dot{L}\right]}{(AL)^2} = \frac{\dot{K}}{AL} - \frac{K}{AL}\left[\frac{\dot{L}A + A\dot{L}}{AL}\right] = \frac{\dot{K}}{AL} - k\left(\frac{\dot{L}}{L} + \frac{\dot{A}}{A}\right).$$

Since factors are paid their marginal product and all income other than labor income is saved, we can write the capital-accumulation equation in this version of the model as

$$(2)\quad \dot{K} = F_K K - \delta K,$$

where F_K is the marginal product of capital. Substituting equation (2) and the constant growth rates of the labor force and technology, $\dot{L}/L = n$ and $\dot{A}/A = g$, into equation (1) yields

$$(3)\quad \dot{k} = \frac{F_K K - \delta K}{AL} - (n+g)k = F_K k - (n+g+\delta)k.$$

The marginal product of capital, F_K, equals $f'(k)$ and so we can rewrite equation (3) as

$$(4)\quad \dot{k} = \left[f'(k) - (n+g+\delta)\right]k.$$

Capital per unit of effective labor will be constant when $\dot{k} = 0$, i.e. when $[f'(k) - (n + g + \delta)]\,k = 0$. This condition holds if $k = 0$ (a case we will ignore) or $f'(k) - (n + g + \delta) = 0$. Thus the balanced-growth-path level of the capital stock per unit of effective labor is implicitly defined by $f'(k^*) = (n + g + \delta)$. Since capital per unit of effective labor, $k \equiv K/AL$, is constant on the balanced growth path, K must grow at the same rate as AL, which grows at rate $n + g$. Since the production function has constant returns to capital and effective labor, which both grow at rate $n + g$ on the balanced growth path, output must also grow at rate $n + g$ on the balanced growth path. Thus we have found a balanced growth path where all the variables of the model grow at constant rates.

The next step is to show that the economy converges to this balanced growth path. At $k = k^*$, $f'(k) = (n + g + \delta)$. If $k > k^*$, $f'(k) < (n + g + \delta)$. This follows from the assumption that $f''(k) < 0$ which means that $f'(k)$ falls as k rises. Thus, if $k > k^*$, we have $\dot{k} < 0$ so that k will fall toward its balanced- growth-path value. If $k < k^*$, $f'(k) > (n + g + \delta)$. Again, this follows from the assumption that $f''(k) < 0$ which means that $f'(k)$ rises as k falls. Thus, if $k < k^*$, we have $\dot{k} > 0$ so that k will rise toward its balanced-growth-path value. Thus, regardless of the initial value of k (if it is not zero), the economy will converge to a balanced growth path at k^*, where all the variables in the model are growing at constant rates.

The golden-rule level of k – the level of k that maximizes consumption per unit of effective labor – is defined implicitly by $f'(k^{GR}) = (n + g + \delta)$. This occurs when the slope of the production function equals the slope of the break-even investment line. Note that this is exactly the level of k that the economy converges to in this model where all labor income is consumed and all other income is saved. Intuitively, saving equals capital's contribution to output, which is the marginal product of capital times the amount of capital. If that contribution exceeds break-even investment, $(n + g + \delta)k$, then k rises. If it is less than break-even investment, k falls. Thus, k settles down to a point where saving, the marginal product of capital times k, equals break-even investment, $(n + g + \delta)k$. That is, the economy settles down to a point where $f'(k)k = (n + g + \delta)k$ or equivalently $f'(k) = (n + g + \delta)$.

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Problem 1.13

We know that $\dot{y}$ is determined by k but since k = g(y), where $g(\bullet) = f^{-1}(\bullet)$, we can write $\dot{y} = \dot{y}(y)$. When k = k* and thus y = y*, $\dot{y} = 0$. A first-order Taylor-series approximation $\dot{y}(y)$ around y = y* of therefore yields

$$(1)\ \dot{y} \cong \left[\frac{\partial \dot{y}}{\partial y}\bigg|_{y=y^*}\right](y - y^*).$$

Let λ denote $-\partial \dot{y}(y)/\partial y\big|_{y=y^*}$. With this definition, equation (1) becomes

$$(2)\ \dot{y}(t) \cong -\lambda[y(t) - y^*].$$

Equation (2) implies that in the vicinity of the balanced growth path, y moves toward y* at a speed

approximately proportional to its distance from y*. That is, the growth rate of $y(t) - y^*$ is approximately constant and equal to $-\lambda$. This implies

(3) $y(t) \cong y^* + e^{-\lambda t}[y(0) - y^*]$,

where y(0) is the initial value of y. We now need to determine λ.

Taking the time derivative of both sides of the production function,

(4) $y = f(k)$,

yields

(5) $\dot{y} = f'(k)\dot{k}$.

The equation of motion for capital is given by

(6) $\dot{k} = sf(k) - (n + g + \delta)k$.

Substituting equation (6) into equation (5) yields

(7) $\dot{y} = f'(k)[sf(k) - (n + g + \delta)k]$.

Equation (7) expresses $\dot{y}$ in terms of k. But $k = g(y)$ where $g(\bullet) = f^{-1}(\bullet)$. Thus we can write

(8) $$\left.\frac{\partial \dot{y}}{\partial y}\right|_{y=y^*} = \left[\left.\frac{\partial \dot{y}}{\partial k}\right|_{y=y^*}\right]\left[\left.\frac{\partial k}{\partial y}\right|_{y=y^*}\right]$$

Taking the derivative of $\dot{y}$ with respect to k gives us

(9) $$\frac{\partial \dot{y}}{\partial k} = f'(k)[sf(k) - (n + g + \delta)k] + f'(k)[sf'(k) - (n + g + \delta)]$$.

On the balanced growth path, $sf(k^*) = (n + g + \delta)k^*$ and thus

(10) $$\left.\frac{\partial \dot{y}}{\partial k}\right|_{y=y^*} = f'(k^*)[sf'(k^*) - (n + g + \delta)]$$.

Now, since $k = g(y)$ where $g(\bullet) = f^{-1}(\bullet)$,

(11) $$\left.\frac{\partial k}{\partial y}\right|_{y=y^*} = \frac{1}{\left.\frac{\partial y}{\partial k}\right|_{y=y^*}} = \frac{1}{f'(k^*)}$$.

Substituting equations (10) and (11) into equation (8) yields

(12) $$\left.\frac{\partial \dot{y}}{\partial y}\right|_{y=y^*} = f'(k^*)[sf'(k^*) - (n + g + \delta)]\frac{1}{f'(k^*)}$$,

or simply

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(13) $$\left.\frac{\partial \dot{y}}{\partial y}\right|_{y=y^*} = sf'(k^*) - (n + g + \delta)$$.

And thus

(14) $$\lambda \equiv -\left.\frac{\partial \dot{y}}{\partial y}\right|_{y=y^*} = (n + g + \delta) - sf'(k^*)$$.

Since $s = (n + g + \delta)k^*/f(k^*)$ on the balanced growth path, we can rewrite (14) as

(15) $$\lambda \equiv -\left.\frac{\partial \dot{y}}{\partial y}\right|_{y=y^*} = (n + g + \delta) - \frac{(n + g + \delta)k^* f'(k^*)}{f(k^*)}$$.

Now use the definition that $\alpha_K \equiv kf'(k)/f(k)$ to rewrite (15) as

(16) $$\lambda \equiv -\left.\frac{\partial \dot{y}}{\partial y}\right|_{y=y^*} = [1 - \alpha_K(k^*)](n + g + \delta)$$.

Thus y converges to its balanced-growth-path value at rate $[1 - \alpha_K(k^*)](n + g + \delta)$, the same rate at

which k converges to its balanced-growth-path value.

Problem 1.14

(a) The production function with capital-augmenting technological progress is given by

(1) $Y(t) = [A(t)K(t)]^{\alpha} L(t)^{1-\alpha}$.

Dividing both sides of equation (1) by $A(t)^{\alpha/(1-\alpha)}L(t)$ yields

(2) $$\frac{Y(t)}{A(t)^{\alpha/(1-\alpha)} L(t)} = \left[\frac{A(t)K(t)}{A(t)^{\alpha/(1-\alpha)} L(t)}\right]^{\alpha} \left[\frac{L(t)}{A(t)^{\alpha/(1-\alpha)} L(t)}\right]^{1-\alpha},$$

and simplifying:

(3) $$\frac{Y(t)}{A(t)^{\alpha/(1-\alpha)} L(t)} = \left[\frac{A(t)^{1-\alpha/(1-\alpha)} K(t)}{L(t)}\right]^{\alpha} A(t)^{-\alpha} = \left[\frac{A(t)^{1-\alpha/(1-\alpha)} A(t)^{-1} K(t)}{L(t)}\right]^{\alpha},$$

and thus finally

(4) $$\frac{Y(t)}{A(t)^{\alpha/(1-\alpha)} L(t)} = \left[\frac{K(t)}{A(t)^{\alpha/(1-\alpha)} L(t)}\right]^{\alpha}.$$

Now, defining $\phi \equiv \alpha/(1 - \alpha)$, $k(t) \equiv K(t)/A(t)^{\phi}L(t)$ and $y(t) \equiv Y(t)/A(t)^{\phi}L(t)$ yields

(5) $y(t) = k(t)^{\alpha}$.

To analyze the dynamics of k(t), take the time derivative of both sides of $k(t) \equiv K(t)/A(t)^{\phi}L(t)$:

(6) $$\dot{k}(t) = \frac{\dot{K}(t)\left[A(t)^{\phi} L(t)\right] - K(t)\left[\phi A(t)^{\phi-1} \dot{A}(t) L(t) + L(t)\dot{A}(t)^{\phi}\right]}{\left[A(t)^{\phi} L(t)\right]^2},$$

(7) $$\dot{k}(t) = \frac{\dot{K}(t)}{A(t)^{\phi} L(t)} - \frac{K(t)}{A(t)^{\phi} L(t)}\left[\phi \frac{\dot{A}(t)}{A(t)} + \frac{\dot{L}(t)}{L(t)}\right],$$

and then using $k(t) \equiv K(t)/A(t)^{\phi}L(t)$, $\dot{A}(t)/A(t) = \mu$ and $\dot{L}(t)/L(t) = n$ yields

(8) $\dot{k}(t) = \dot{K}(t)/A(t)^{\phi} L(t) - (\phi\mu + n)k(t)$.

The evolution of the total capital stock is given by the usual

(9) $\dot{K}(t) = sY(t) - \delta K(t)$.

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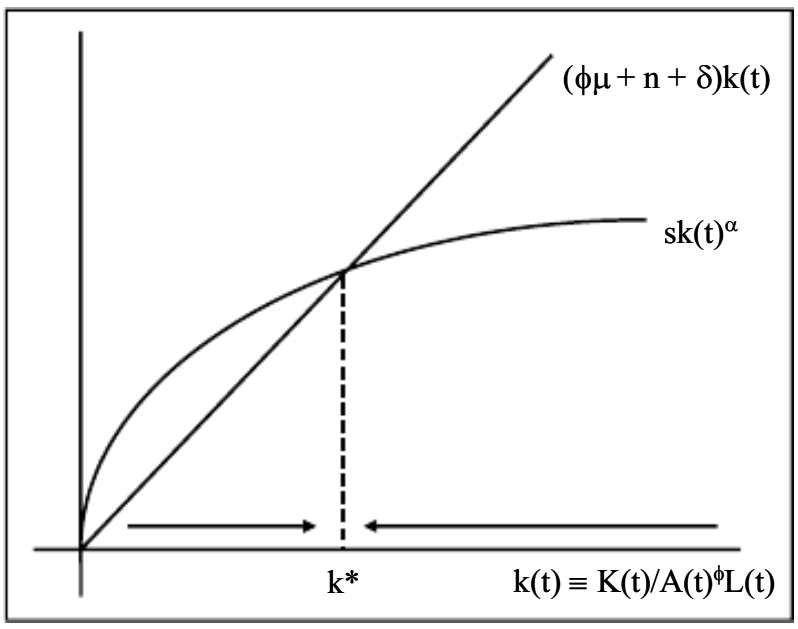
Substituting equation (9) into (8) gives us

(10) $\dot{k}(t) = sY(t)/A(t)^{\phi}L(t) - \delta K(t)/A(t)^{\phi}L(t) - (\phi\mu + n)k(t) = sy(t) - (\phi\mu + n + \delta)k(t).$

Finally, using equation (5), $y(t) = k(t)^{\alpha}$, we have

(11) $\dot{k}(t) = sk(t)^{\alpha} - (\phi\mu + n + \delta)k(t).$



Equation (11) is very similar to the basic equation governing the dynamics of the Solow model with labor-augmenting technological progress. Here, however, we are measuring in units of $A(t)^{\phi}L(t)$ rather than in units of effective labor, $A(t)L(t)$. Using the same graphical technique as with the basic Solow model, we can graph both components of $\dot{k}(t)$. See the figure at right.

When actual investment per unit of $A(t)^{\phi}L(t)$, $sk(t)^{\alpha}$, exceeds break-even investment per unit of $A(t)^{\phi}L(t)$, given by $(\phi\mu + n + \delta)k(t)$, k will rise toward k*. When actual investment per unit of $A(t)^{\phi}L(t)$ falls short of break-even investment per unit of $A(t)^{\phi}L(t)$, k will fall toward k*. Ignoring the case in which the initial level of k is zero, the economy will converge to a situation in which k is constant at k*. Since $y = k^{\alpha}$, y will also be constant when the economy converges to k*.

The total capital stock, K, can be written as $A^{\phi}Lk$. Thus when k is constant, K will be growing at the constant rate of $\phi\mu + n$. Similarly, total output, Y, can be written as $A^{\phi}Ly$. Thus when y is constant, output grows at the constant rate of $\phi\mu + n$ as well. Since L and A grow at constant rates by assumption,

we have found a balanced growth path where all the variables of the model grow at constant rates.

(b) The production function is now given by

(12) $Y(t) = J(t)^{\alpha} L(t)^{1-\alpha}$.

Define $\bar{J}(t) \equiv J(t)/A(t)$. The production function can then be written as

(13) $Y(t) = \left[A(t)\bar{J}(t)\right]^{\alpha} L(t)^{1-\alpha}$.

Proceed as in part (a). Divide both sides of equation (13) by $A(t)^{\alpha/(1-\alpha)}L(t)$ and simplify to obtain

(14) $$\frac{Y(t)}{A(t)^{\alpha/(1-\alpha)} L(t)} = \left[\frac{\bar{J}(t)}{A(t)^{\alpha/(1-\alpha)} L(t)}\right]^{\alpha}.$$

Now, defining $\phi \equiv \alpha/(1 - \alpha)$, $\bar{j}(t) \equiv \bar{J}(t)/A(t)^{\phi}L(t)$ and $y(t) \equiv Y(t)/A(t)^{\phi}L(t)$ yields

(15) $y(t) = \bar{j}(t)^{\alpha}$.

In order to analyze the dynamics of $\bar{j}(t)$, take the time derivative of both sides of $\bar{j}(t) \equiv \bar{J}(t)/A(t)^{\phi}L(t)$:

(16) $$\dot{\bar{j}} = \frac{\dot{\bar{J}}(t)\left[A(t)^{\phi}L(t)\right] - \bar{J}(t)\left[\phi A(t)^{\phi-1}\dot{A}(t)L(t) + L(t)A(t)^{\phi}\dot{L}(t)\right]}{\left[A(t)^{\phi}L(t)\right]^2},$$

(17) $$\dot{\bar{j}}(t) = \frac{\dot{\bar{J}}(t)}{A(t)^{\phi}L(t)} - \frac{\bar{J}(t)}{A(t)^{\phi}L(t)}\left[\phi\frac{\dot{A}(t)}{A(t)} + \frac{\dot{L}(t)}{L(t)}\right],$$

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and then using $\bar{j}(t) \equiv \bar{J}(t)/A(t)^{\phi}L(t)$, $\dot{A}(t)/A(t) = \mu$ and $\dot{L}(t)/L(t) = n$ yields

(18) $\dot{\bar{j}}(t) = \dot{\bar{J}}(t)/A(t)^{\phi}L(t) - (\phi\mu + n)\bar{j}(t)$.

The next step is to get an expression for $\dot{\bar{J}}(t)$. Take the time derivative of both sides of $\bar{J}(t) \equiv J(t)/A(t)$:

(19) $$\dot{\bar{J}}(t) = \frac{\dot{J}(t)A(t) - J(t)\dot{A}(t)}{A(t)^2} = \frac{\dot{J}(t)}{A(t)} - \frac{\dot{A}(t)}{A(t)}\frac{J(t)}{A(t)}.$$

Now use $\bar{J}(t) \equiv J(t)/A(t)$, $\dot{A}(t)/A(t) = \mu$ and $\dot{J}(t) = sA(t)Y(t) - \delta J(t)$ to obtain

(20) $$\dot{\bar{J}}(t) = \frac{sA(t)Y(t)}{A(t)} - \frac{\delta J(t)}{A(t)} - \mu\bar{J}(t),$$

or simply

(21) $\dot{\bar{J}}(t) = sY(t) - (\mu + \delta)\bar{J}(t)$.

Substitute equation (21) into equation (18):

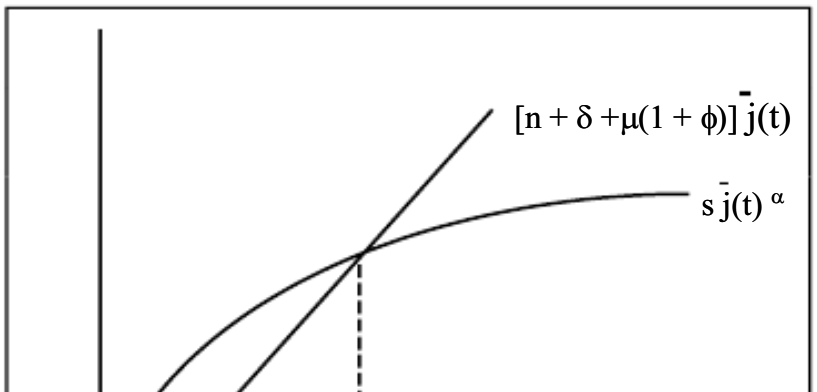
(22) $\dot{\bar{j}}(t) = sY(t)/A(t)^{\phi}L(t) - (\mu + \delta)\bar{J}(t)/A(t)^{\phi}L(t) - (\phi\mu + n)\bar{j}(t) = sy(t) - \left[n + \delta + \mu(1+\phi)\right]\bar{j}(t)$.

Finally, using equation (15), $y(t) = \bar{j}(t)^{\alpha}$, we have

(23) $\dot{\bar{j}}(t) = s\bar{j}(t)^{\alpha} - \left[n + \delta + \mu(1+\phi)\right]\bar{j}(t)$.

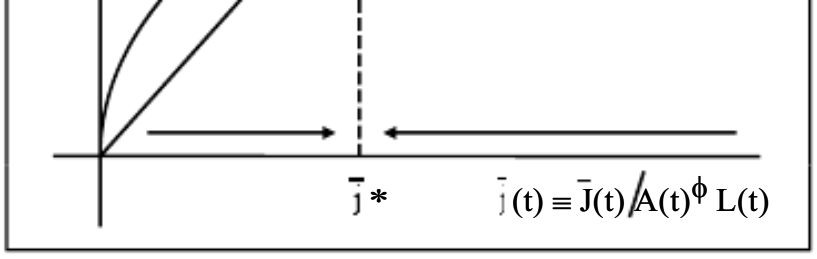
Using the same graphical technique as in the basic Solow model, we can graph both components of $\dot{\bar{j}}(t)$.

See the figure at right. Ignoring the possibility that the initial value of $\bar{j}$ is zero, the economy will converge to a situation where $\bar{j}$ is constant at $\bar{j}^*$. Since $y = \bar{j}^{\alpha}$, y will also be constant when the economy converges to $\bar{j}^*$.



The level of total output, Y, can be written as $A^{\phi}Ly$. Thus when y is

constant, output grows at the constant rate of $\phi\mu + n$.



By definition, $\bar{J} \equiv A^{\phi}\bar{L}j$. Once the economy converges to the situation where $\bar{j}$ is constant, $\bar{J}$ grows at the constant rate of $\phi\mu + n$. Since $J \equiv \bar{J}\bar{A}$, the effective capital stock, J, grows at rate $\phi\mu + n + \mu$ or $n + \mu(1 + \phi)$. Thus the economy does converge to a balanced growth path where all the variables of the model are growing at constant rates.

(c) On the balanced growth path, $\dot{\bar{j}}(t) = 0$ and thus from equation (23):

(24) $s\bar{j}^{\alpha} = [n + \delta + \mu(1+\phi)]\bar{j} \quad \Rightarrow \quad \bar{j}^{1-\alpha} = s/[n + \delta + \mu(1+\phi)]$,

and thus

(25) $\bar{j}^* = [s/(n + \delta + \mu(1+\phi))]^{1/(1-\alpha)}$.

Substitute equation (25) into equation (15) to get an expression for output per unit of $A(t)^{\phi}L(t)$ on the balanced growth path:

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